LAB-1. Demonstrate various data pre-processing technique for a given dataset.

Q) Write python Code, consider filename as 'housing.csv'

- To load .csv file into the dataframe
- To display information of all columns
- To display statistical information of all numerical
- To display the count of unique labels for 'Ocean proximity' column
- To display which attribute in a dataset have missing value count greater than 0

⇒
```
import pandas as pd
df = pd.read.csv ("housing.csv")
print (df.info())
Print (df.describe())
Print (df ["Ocean proximity"].value count ())
miss-values = df.isnull() sum()
print (miss-values [miss-values>0])
```

for both datasets ⇒

1. Which column in the dataset had missing values? How did you handle them?

2. Which categorial column did you ~~identy~~ identify in the dataset? How did you encode them?

3. What is the differente between min-max scaling and standaudisation when would you use one over the other.

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Diabetic.

These were no missing values in any columns.

Adults

Contains missing values (?) in column like workclass, occupation native country

We replaced

df[numerical_cols] = df[numerical_cold fillna(
df[numerical_cols]. mean())

② Diabetes

Gender and class

Encoding nutho ds =)

le = label Encoder()

df[col] = le. fit. transform (dF[col])

Adult

workclass ~~education~~ ~~the~~ education,

mental state, occupation

encoding method

for col. in df. select_dtype(include = 'object'). columns:

df[col] = le. fit_transform (dF[col])

min-max

Scale value range b/w 0 & 1

formula: x scaled = $\frac{x - x_{min}}{x_{max} - x_{min}}$

used for neural networks, KNN

Standardisation

L scale value Range to 0-mean & unit variance

L formula: ~~x~~ x scaled = $\frac{x - \mu}{\sigma}$

1. Perform the

Describe () and info()

- info() Shows data types and missing values.
- describe() shows mean, min, max and spread of numerical data.

2. Histograms.

- Show distribution of each feature
- median - income is right skewed.
- Housing - median - age is capped at 52.

3. Random vs Stratified Test set -

- Random: Selects data randomly
- Stratified: keeps same income proportion in train & test

4. Geographical features.

longitude and latitude.

- Coastal areas have higher price.

5 Correlation

- show relationship with house price.
- Highest correlation: median - income.

6. Combined features.

- Create features like rooms per household

7. Data Cleaning

- total - bedrooms has missing values.
- Filled using median

8 Categorical to Numerical

- Ocean - proximity is categorical
- used one - Hot - encoding

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9. Features Scaling

- Makes all features on same scale
- important for distance-based models

10. Pipeline

: combines cleaning, features creation, scaling, and encoding

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Lab 13

Date 9

Demonstrate the steps to build a machine-learning model that predicts the median housing price using the California housing price dataset

1 perform the describe and info steps

housing.info()

housing.describe()

2. Plot the histogram of each feature. Indicate what histogram indicates on median-incomes & house-median-age.

→ import matplotlib.pyplot as plt

housing.hist(bin=50, figsize=(15,10))

plt.show()

→ What histogram indicate.

1) median income

- The distribution is right-skewed (+ve)
- most valus betn 2&6
- few districts house income above 200
- Extreme value near 15

=> Since it is skewed, scaling is important before modelling

ii) Housing-median-age

- Distribution is spread between 1 and 50
- Noticeable spike at 52

=> Ciling effect seen.

=> Distribution more uniform compact to median income

3. Demonstrate pro

Random test set	Stratified Test set
· Data is split randomly into training & testing set	· Data is split while maintaining proportion of important categories
· Does not guarantee distribution of imp features.	· Ensures test set represents overall dataset properly
· May cause sampling bias if dataset is skewed.	· Reduces sampling bias
· Simple & quick.	· more reliable for imbalanced or skewed datasets

4)

→ Geographical features : latitude, Longitude
→ What graph indicates
· higher price near costal area
· lower price in inland area
· population effect (high popⁿ ↔ hig price.

5

median-house-value correlation maximum
median-income v How price.
graph shows · clear upward trend
· higher income → higher house pr
· some price cap at 500000

6.

→ total-room ÷ household = rooms-per-household
total-bedroom ÷ total room = bedroome-per-room
population ÷ household = pop-per-household

7

→ Total . bedroom
housing-num = housing. drop ("ocean-proximity", axis=1)
impute fit(housing-num)
housing-num-new = impute. transform (housing-num)

8

→ category col : ocean proximity
~~categories ? it to explain method used to count & demonstrate.~~

categories : < 1Hocean , Inland , near ocean , near bay , island

method ① One Hot Encoding
Convert each category into seperate binary (0, 1) column

② label encoding
assign no's like inland = 0 nearbay = 1

method used → Ordend date

oceanproximity	inland	nearbay	near ocean	<1Hoc	island
inland	1	0	0	0	0
near bay	0	1	0	0	0
near Ocean	0	0	1	0	0
<1H ocean	0	0	0	1	0
island	0	0	0	0	1

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9

→ feature scaling is process of bringing all numerical feature to a similar range or scale so that no feature dominate other due to range magnitude

Importance

- Prevents feature dominance
- Improve model performance
- ~~faster~~ Convergences.
- Required for regularization

10.

→ Simple impute : fills missing values using media

→ custom transformer : create new meaningful features, improves corr

→ Strandard scale : Scales numerical features to mea

→ One Hot Encode : categorical var to binary col

→ Column transformer : Applies numerical pipeline to numeric column

Applies Encoding to categorical col

Combines all process & features in one df.

Logistic Regression.

1) Binary classification

Given

$x_0 = -5$, $x_1 = 0.8$

a) Equation

$$P(y=1) \; \frac{1}{1+e^{-(-5+0.8x)}}$$

b) For 7 hours

$$z = -5 + (0.8 \times 7) = 0.6$$

$$P = \frac{1}{1+e^{-0.6}} \approx 0.645$$

Probability $= 0.645$ (64.5%)

c) Predicted class

Since $0.645 > 0.5$

~~Pass~~

2) Softmax

Given $z = [2, 1, 0]$

$e^2 = 7.389$

$e^1 = 2.718$

$e^0 = 1$

Sum $= 11.107$

Probabilities:

Class 1 $= 0.665$ $= 7.389 / 11.107$

Class 2 $= 0.245$ $= 2.718 / 11.107$

Class 3 $= 0.090$ $= 1 / 11.107$ /1.

HR Dataset observation

Important Variables

- ~~Statis~~ Satisfaction level
- Salary
- promotion in last 5 year
- project
- monthly hours

These directly affect employee retention

ii) Accuracy

Accuracy = (write your value, eg 78%)

This is good accuracy because most employe were predicted correctly

200 Dataset obsefvation.

1) pre processing
- Removed name column.
- Encoded Categorical value.
- checked missing value.

ii) missing value
No major missing value found.

iii) Confusion matrix
Diagonal values show correct predictions
Higher diagonal = better performance.

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iv) misclassified class :

Some similar animal type were confused due to similar feature.

Decision Tree

IRIS Dataset

Accuracy (write your value, eg. 95%)

Confusion matrix

most prediction correct.

Some confusion betn versicolor and virginica

~~Drug~~ Dataset

Accuracy ≈ ~~[illegible]~~ 93%.

Few misclassification between similar drug types

Petrol Consumption (Regression Tree)

Errors

MAE = 35.4

MSE = 1800.6

RMSE = 42.43

Interpretation

Important features:

- petrol Tax
- Average Income
- population
- Highway length.

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Regression Tree predicts continuous values and minimizing Squared error, unlike Classification tree which uses entropy or Gini index.

24/2/26

Test Data = (x, 35, 100)
k = 3

Training Dataset

Person	Age	Salary	Target
A	25	40	No
B	35	60	No
C	45	80	Yes
D	20	20	No
E	35	120	Yes

Step 1. Distance formula

$$d = \sqrt{(Age_1 - Age_2)^2 + (Salary_1 - Salary_2)^2}$$

Test point = (35, 100)

Step: 2 : Calculate distance A(25, 40)

$$d = \sqrt{(35-25)^2 + (100-40)^2}$$
$$= 60.83$$

Distance from B (35, 60)

$$d = \sqrt{(35-35)^2 + (100-60)^2}$$
$$= 40$$

Distance from C (45, 80)

$$\sqrt{(35-45)^2 + (100-80)^2}$$
$$= 22.36$$

Distance from D (20,20)

$$d = \sqrt{(35-20)^2 + (100-20)^2}$$
$$= 81.39$$

Distance from E (35,120)

$$d = \sqrt{(35-35)^2 + (100-120)^2}$$
$$= 20$$

Sort Distance.

Person	Distance	Target
E	20	Yes
C	22.36	Yes
B	40	No
A	60.83	No
D	81.39	No

Select k = 3 Nearest Neighbours.

person	Target
E	Yes
C	Yes
B	No

Step 5 : Majority Voting

Yes = 2

No = 1

Majority class = Yes

Final prediction

Test Data (x, 25, 100) → Target = Yes

Result : Using kNN classifier with k = 3, the predicted class for person x is Yes.

1. Iris Dataset

How to choose the k value ?

To choose the best k value, test different values of k and calculate Accuracy rate and Error Rate

- Accuracy Rate = Correct prediction / Total prediction
- Error Rate = 1 - Accuracy

Example :

k.	Accuracy	Error
1	0.90	0.10
3	0.94	0.06
5	0.96	0.04

Choose the k value with higher accuracy and lowest error (k=5)

Diabetes Dataset

Purpose of feature Scaling

Features Scaling is used to normalize data values so all features have the same scale.

This helps KNN calculate distances correctly and improves accuracy

How to perform it

```
from sklearn.preprocessing import standard scaler
Scaler = standard scaler()
X_train = ~~standar~~ scaler.fit_transform (X_train
X_test = scaler.transform (X_test.) .
```

5/1/26

Q. 1

Points (4,1), (4,-1) and (6,0) belong to positive class and points (1,0), (0,1) and (0,-1) belongs to negative class. Draw an optimal hyperplane.

Given points

Positive class: (4,1), (4,-1), (6,0)

Negative class (-1): (1,0), (0,1), (0,-1)

Step 1: Observation the data.

- All negative points are around $x=0$ and $x=1$
- All positive points are around $x=4$ and $x=6$.

So the classes are mainly separated along the x-axis.

Step 2: Find the closest point between class

The closest points from each class are;

- Negative: (1,0)
- Positive: (4,1) and (4,-1) ($x=4$)

For maximum margin, the hyperplane should lie midway between $x=1$ and $x=4$

$$x = \frac{1+4}{2} = 2.5$$

Step 3: Optimal Hyperplane.

The optimal separating hyperplane is

$$x = 2.5$$

or in the form

$$w \cdot x + b = 0$$

$$x - 2.5 = 0$$

where

$w = (1,0)$

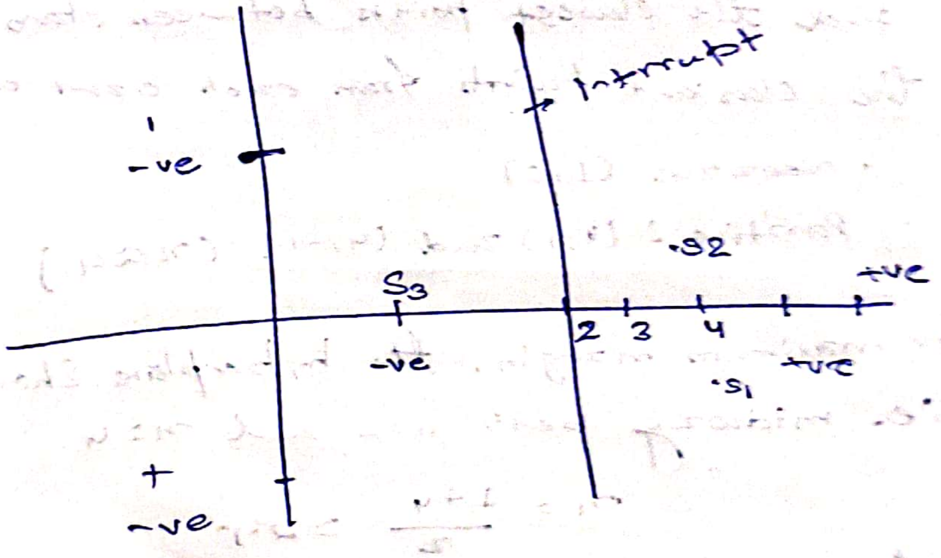
$b = -2.5$

Step 4: Support vector

Support vectors are the closest points to the hyperplan

- (1,0) from negative class
- (4,1) and (4,-1) from +ve class

Optimal Hyperplane : $x = 2.5$



For Iris.csv dataset

Accuracy Score

- linear kernel Accuracy - 96% - 97%.
- RBF kernel Accuracy : 98 - 100%.

Exact value may vary depending on dataset split

Which kernel perform better?

- RBF kernel performed slightly better than the linear kernel

Reason.

The RBF kernel can handle non-linear decision boundaries while the linear kernel only separate data with a straight line.

2. For letter-recognition.csv.dataset

Confusion Matrix Interpretation

The confusion matrix show ~~correct~~ the number of correct and incorrect prediction for each letter class

- O and Q
- I and L
- C and G

This happens because the feature describing those letter are very similar

AUC (Area Under curve) Score

AUC measures how well the classifier distinguishes btn classes

- AUC value close to 1 → Excellent classifier
- AUC value close to 0.5 → Poor classifier

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In this dataset the AUC score is usually high (around 0.9 or above) showing good classification performance

Performance comparison with IRIS dataset

∴ The IRIS dataset is smaller and simpler. So the SVM model achieves high accuracy

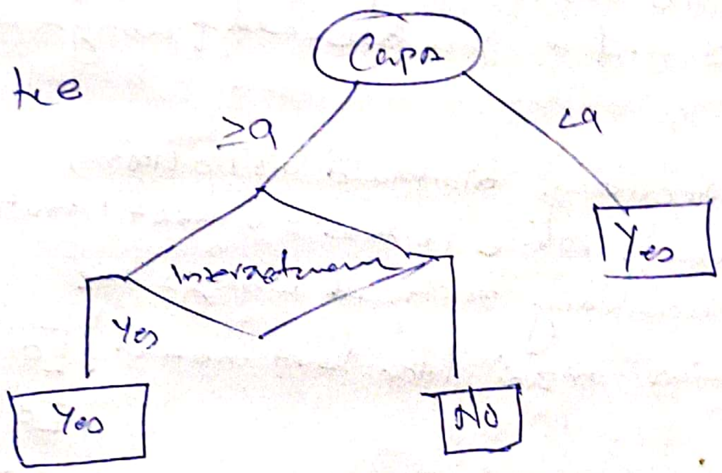
Lab-7

Implement Random Forest ensembly method on given datasets

Q. For S1 shown the decision tree Considering CGPA as a root node

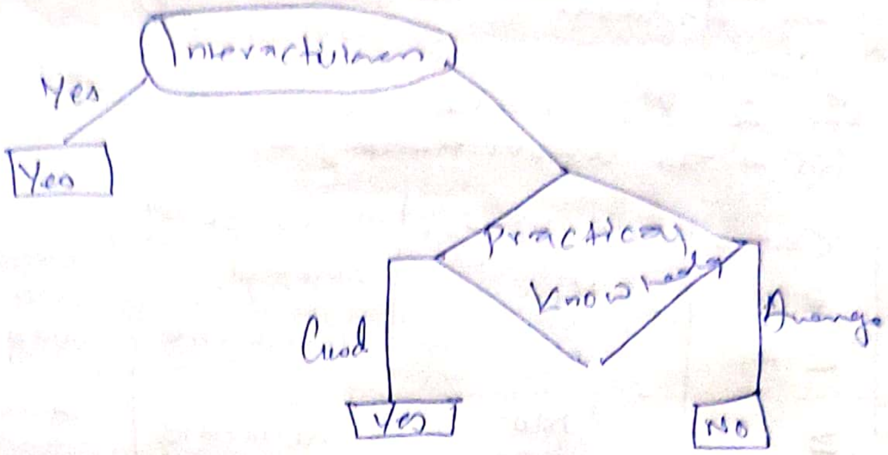
Sno	CGPA	interactiveness	Commonskills	practical knowledge	Job offer
1	≥ 9	Yes	Good	Good	Yes
2	< 9	No	Moderate	Good	Yes
3	≥ 9	No	Moderate	Avg	No
4	≥ 9	No	Moderate	Avg	No
5	≥ 9	Yes	Moderate	Good	Yes

Decision tree



Q For Sample S2 shown, draw the decision tree Considering interactiveness as root node

Sno	CGPA	interactiveness	Common skills	Prac know	Job offer
1	<9	No	moderate	Good	Yes
2	≥9	No	moderate	Avg	No
3	≥9	No	moderate	Avg	No
4	≥9	Yes	Moderate	Good	Yes
5	≥9	Yes	Moderate	Good	Yes



Iris dataset

Q what is the best accuracy score and confusion matrix of the classifier you obsurved & using how many trees

→ Best accuracy obtained is 1.0 (100%)
Accuracy while 10 tree = 0.9667 (96.67 %)
Best accuracy achived using 21 tree
Confusion matrix (for best model) [9 0 0]
[0 9 0]
[0 0 12]

for default =10 tree [9 0 0]
[0 9 0]
[0 1 1]

Implement Adaboosting ensemble method.

Q. Consider Adaboost algorithms for the following sample data, Show the Decision stump calculation step for the attribute CGPA.

CGPA	Interactiveness	Practical knowledge	Common skills	Job
>=9	yes	Good	Good	yes
<9	NO	Good	moderate	yes
>=9	No	Avg	moderate	No
<9	No	Avg	good	No
>=9	yes	good	moderate	yes
>=9	yes	Good	moderate	yes

Step
(a) random bootstrap Sample

CGPA >=9 → yes
<9 → NO

CGPA	prediction Job offer	Actual Job	weight
>=9	Yes	Yes	1/6
<9	No	yes	1/6
>=9	Yes	No	1/6
<9	NO	No	1/6
>=9	yes	yes	1/6

Step

(b) $\varepsilon_i = \sum_{j=1}^{N} H_i(d_j) wt(d_j)$

where $H_i(d_i) = 0$ if prediction correct with H_i

$H_i(d_j) = 1$ prediction worse

$$\varepsilon_{CGPA} = (1)\left(\frac{1}{6}\right) + (1)\left(\frac{1}{6}\right) = \frac{1}{3} = 0.333$$

Step 2 Calculate weight of weak classifier

$$\alpha_{CGPA} = \frac{1}{2} \ln\left(\frac{1-\varepsilon_{CGPA}}{\varepsilon_{CGPA}}\right) = \frac{1}{2} \times \ln\frac{(1-0.333)}{0.333} = 0.34$$

Step (d) calculate normalizing factor

$$Z_{CGPA} = \left(\tfrac{1}{6} \times 4\right) + \left(\tfrac{1}{6} \times 2 \times e^{0.347}\right)$$

$= 0.9428$

update weight

$$w_t(d_i)_{t+1} = \frac{w_t(d_j)_{\alpha_{PA}} \text{ of correct instan } \times e^{-\alpha_{PA}}}{Z_{\alpha_{PA}}}$$

$$= \frac{(1/6)\, e^{-0.347}}{6.9428}$$

$= 0.1249$ ← correct instance

* income.csv

What is the best accuracy score & confusion matrix of classification & using how many trees?

→ Accuracy with 10 estimates : 81.82%
Best accuracy ⇒ 83.40% → 73

Confusion matrix

(10) [[6782 632]
[1144 1211]]

Confusion matrix

(73) [[7705 599]
[1223 1132]]

k-mean.

Q Cluster the following into 3 cluster

A1 (2,10), A2 (2,5), A3 (8,4), A4 (5,8)

A5 (7,5) A6 (6,4) A7 (1,2) A8 (4,9)

Initial cluster centre are : A1 (2,10)
A4 (5,8)
A7 (1,2)

iteration 1

Point	(2,10) C1	(5,8) C2	(1,2) C3	Cluster
(2,10)	6	5	9	C1
(2,5)	5	2	4	C3
(8,4)	12	7	9	C2
(5,8)	5	0	10	C2
(7,5)	10	5	9	C2
(6,4)	10	5	7	C2
(1,2)	9	10	0	C3
(4,9)	3	2	10	C2

new centers

C1 : A1 (2,10)

$$C2 : \left(\frac{8+5+7+4+6}{5}\right), \left(\frac{4+8+5+4+9}{5}\right) = (6,6)$$

$$C3 : \left(\left(\frac{2+1}{2}\right), \frac{(5+2)}{2}\right) = (1.5, 3.5)$$

* Implementation Iris.csv

Silhouette score = 0.47872

* Implement dimensionality reduction using PCA method

Q In given table reduce dimension from 2 to 1 using PCA. Complete 1st principle Component

Feature	Example 1	Example 2	Example 3	Example 4
x_1	4	8	13	7
x_2	11	4	5	14

Given Eigen value

$\lambda_1 = 30.3849$

$\lambda_2 = 6.0151$

Eigen vector

$e_1 \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$ $\quad C_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$

Soln

calculate mean

$\bar{x}_1 = \frac{4+8+13+7}{4} = 8$ $\qquad \bar{x}_2 = \frac{11+4+5+14}{4} = 8.5$

covariance matrix

$$cov(x_1 x_1) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{1k} - \bar{x}_1)(x_{1k} - \bar{x}_1)$$

$$= \frac{1}{3}((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2)$$

$$= 14$$

$$Cov(x_1 x_2) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{1k} - \bar{x}_1)(x_{2k} - \bar{x}_2)$$

$$= \frac{1}{3}((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5))$$

$$= -11$$

$$S = \begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) \\ cov(x_2, x_1) & cov(x_2, x_2) \end{bmatrix} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Eigen values:

$$e_1 \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix} \qquad e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

first principle component

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} x_1 - \bar{x}_1 \\ x_{21} - \bar{x}_2 \end{bmatrix}$$

$$= 0.5574(4-8) - 0.8303(11-8.5)$$

$$= -4.30535$$

*

Feature	Ex1	Ex2	Ex3	Ex4
x_1	4	8	13	7
x_2	11	4	5	14
First principle	-4.3052	3.7361	5.6929	-5.1238

* Heart.csv

Repeat accuracy score before & after applying PCA

Before PCA

SVM: 86.4%

logistic regression: 85.32%

Random 87.5%

Best accuracy = 87.5%

After PCA

SVM 85.86%

LR 85.32%

Random for = 85.87%

best